

Fortinet Security Fabric

FortiAnalyzer can recognize a Security Fabric group of devices and display all units in the group on the *Device Manager* pane. See [Adding a Security Fabric group on page 208](#). FortiAnalyzer supports the Security Fabric by storing and analyzing the logs from the units in a Security Fabric group as if the logs are from a single device. You can also view the logging topology of all units in the Security Fabric group for additional visibility. See [Displaying Security Fabric topology on page 209](#).

FortiAnalyzer provides dynamic data and metadata exchange with the Security Fabric and uses the data in FortiView and Reports for additional visibility. A default report template lets you monitor new users, devices, applications, vulnerabilities, threats and so on from the Security Fabric.

A set of dashboard widgets lets you review audit scores for a FortiGate Security Fabric group with recommended best practices and historical audit scores and trends.

If FortiClient is installed on endpoints for endpoint control with FortiGate, you can use the endpoint telemetry data collected by the Security Fabric agent to display user profile photos in reports and FortiView.

Adding a Security Fabric group

Before you can add a Security Fabric group to FortiAnalyzer, you need to create the Security Fabric group in FortiGate.

Fortinet recommends using a dedicated Super_User administrator account on the FortiGate for FortiAnalyzer access. This ensures that associated log messages are identified as originating from FortiAnalyzer activity. This dedicated Super_User administrator account only needs *Read Only* access to *System Configuration*; all other access can be set to *None*.

Once a Security Fabric group is added in FortiAnalyzer, it can be used to filter other panes, such as *Log View* and *FortiView*. You can select the Security Fabric or individual devices in the fabric from the *Device* filter in those panes.

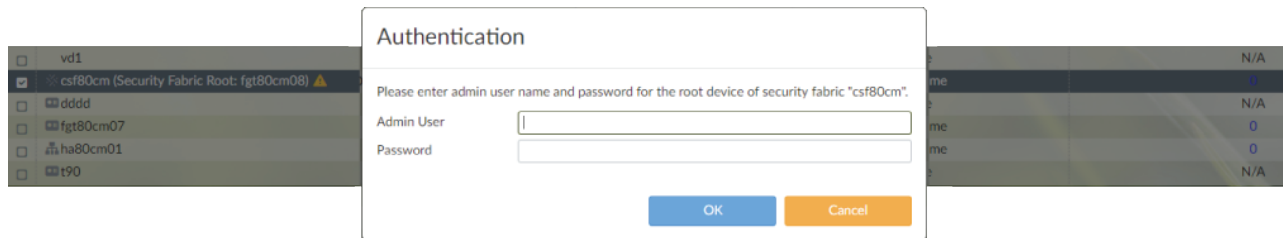
To add a Security Fabric group:

1. Go to *Device Manager > Unauthorized Devices*.
2. Select all the devices corresponding to the Security Fabric group created in FortiGate.



All devices in the Security Fabric group must be added to FortiAnalyzer, and they must be added in the same ADOM.

3. Authenticate the Security Fabric group by clicking the *Warning* icon (yellow triangle) beside the corresponding FortiGate root.



4. Enter the *Authentication Credentials*. The authentication credentials are the ones you specified in FortiGate. Once the FortiGate root has been authenticated, the *Warning* icon will disappear.
5. After authentication, it takes a few minutes for FortiAnalyzer to automatically populate the devices under the FortiGate root which creates the Security Fabric group.

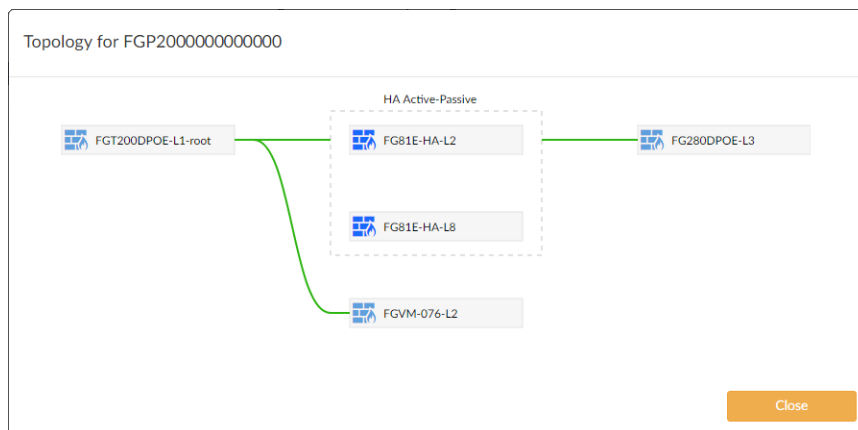
Displaying Security Fabric topology

For Security Fabric devices, you can display the Security Fabric topology.

To display the Security Fabric topology:

1. If using ADOMs, ensure that you are in the correct ADOM.
2. Go to *Device Manager*.
3. Right-click a Security Fabric device and select *Fabric Topology*.
A pop-up window displays the Security Fabric topology for that device.

If you selected *Fabric Topology* by right-clicking a device within the Security Fabric group, the device is highlighted in the topology. If you selected *Fabric Topology* by right-clicking the name of the Security Fabric group, no device is highlighted in the topology.



Security Fabric traffic log to UTM log correlation

FortiAnalyzer correlates traffic logs to corresponding UTM logs so that it can report sessions/bandwidth together with its UTM threats. Within a single FortiGate, the correlation is performed by grouping logs with the same session IDs, source and destination IP addresses, and source and destination ports.

In a Cooperative Security Fabric (CSF), the traffic log is generated by the ingress FortiGate, while UTM inspection (and subsequent logs) can occur on any of the FortiGates. This means that the traffic logs did not have UTM related log fields, as they would on a single FortiGate. Different CSF members also have different session IDs, and NAT can hide or change the original source and destination IP addresses. Consequently, without a proper UTM reference, the FortiAnalyzer will fail to report UTM threats associated with the traffic.

This feature adds extensions to traffic and UTM logs so that they can be correlated across different FortiGates within the same security fabric. It creates a UTM reference across CSF members and generates the missing UTM related log fields in the traffic logs as if the UTM was inspected on a single FortiGate.

NAT translation is also considered when searching sources and destinations in both traffic and UTM logs. The FortiGate will generate a special traffic log to indicate the NAT IP addresses to the FortiAnalyzer within the CSF.

Traffic logs to DNS and SSH UTM references are also implemented - the DNS and SSH counts in Log View can now be clicked on to open the related DNS and SSH UTM log. IPS logs in the UTM reference are processed for both their sources and destinations in the same order, and in the reverse order as the traffic log. The FortiGate log version indicator is expanded and used to make a correct search for related IPS logs for a traffic log.

This feature requires no special configuration. The FortiAnalyzer will check the traffic and UTM logs for all FortiGates that are in the same CSF cluster and create the UTM references between them.

To view the logs:

1. On the FortiAnalyzer, go to *Log View > FortiGate > Traffic*.
The UTM security event list, showing all related UTM events that can happen in another CSF member, is shown.
2. Click the count beside a UTM event to open the related UTM event log window. In this example, the traffic log is from the CSF child FortiGate, and the UTM log is from the CSF root FortiGate.

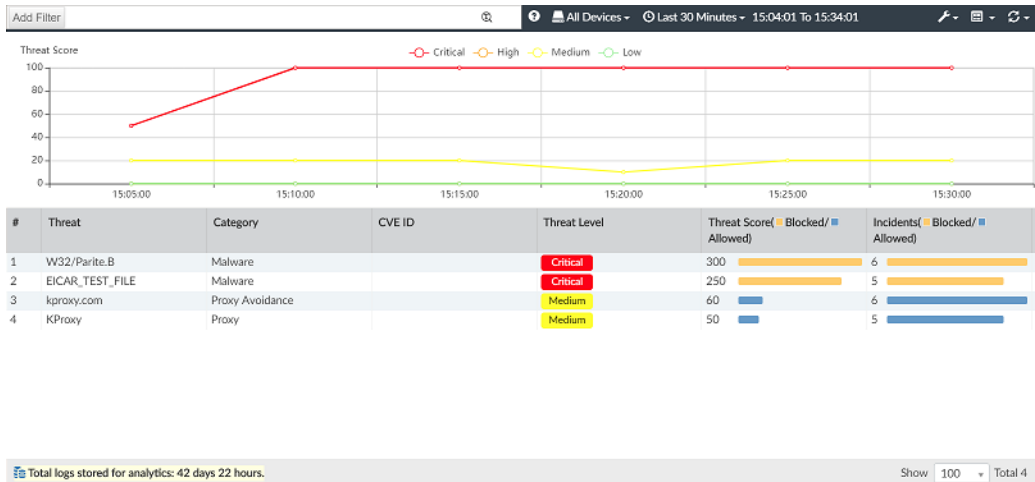
Device Name	Source Port	Action	Security Event List	Application Category	Application Control List	Application ID	Application Risk	Host Name
FGT_61E_CSF_child	33781	Malware	APP 1 AV 1 WEB 1 WAF 2	Web Client	default	15893	medium	172.18.32.126
FGT_61E_CSF_child	42573	✓	APP 1 WEB 1 WAF 1	Proxy	default	16604	critical	kproxy.com
FGT_61E_CSF_child	33779	✓	APP 1 WEB 1 WAF 2	Web Client	default	15893	medium	172.18.32.126
FGT_61E_CSF_child	50053	✓	DNS 2	unscanned				kproxy.com
FGT_61E_CSF_child	57617	✓	DNS 2	unscanned				kproxy.com
FGT_61E_CSF_child	50196	✓	DNS 2	unscanned				kproxy.com
FGT_61E_CSF_child	33778	Malware	APP 1	unscanned				kproxy.com
FGT_61E_CSF_child	35511	Malware	APP 1 AV 1 WEB 1 WAF 2	Antivirus				
FGT_61E_CSF_child	49144	✓	APP 1 WEB 1 WAF 1					
FGT_61E_CSF_child	35510	✓	APP 1 WEB 1 WAF 2					
FGT_61E_CSF_child	37840	✓	DNS 2					
FGT_61E_CSF_child	36613	✓	DNS 2					
FGT_61E_CSF_child	54061	✓	DNS 2					
FGT_61E_CSF_child	33508	✓	DNS 2					
FGT_61E_CSF_child	57952	Malware	APP 1 AV 1 WEB 1 WAF 2					
FGT_61E_CSF_child	49284	✓	APP 1 WEB 1 WAF 1					
FGT_61E_CSF_child	57950	✓	APP 1 WEB 1 WAF 2					
FGT_61E_CSF_child	35400	✓	DNS 2					
FGT_61E_CSF_child	39090	✓	DNS 2					
FGT_61E_CSF_child	47483	✓	DNS 2					
FGT_61E_CSF_child	57949	Malware	APP 1 AV 1 WEB 1 WAF 1					
FGT_61E_CSF_child	35943	✓	APP 1 WEB 1 WAF 2					
FGT_61E_CSF_child	35941	✓	APP 1 WEB 1 WAF 2					
FGT_61E_CSF_child	41376	✓	APP 1 WEB 1 WAF 1					
FGT_61E_CSF_child	48194	✓	DNS 2					
FGT_61E_CSF_child	55391	✓	DNS 2					
FGT_61E_CSF_child	50109	✓	DNS 2					
FGT_61E_CSF_child	35940	Malware	APP 1 AV 1 WEB 1 WAF 1					
FGT_61E_CSF_child	58921	✓	APP 1 WEB 1 WAF 2					

▲ #	Date/Time	Device Name	Policy ID	Action	Source
1	15:16:36	FGT_100D_CSF_root	2	blocked	10.1.10.12

Level	warning
Threat Level	critical
Threat Score	50
Source	
Agent	curl%2F7.19.7
Device ID	FG100D3G16840718
Device Name	FGT_100D_CSF_root
End User ID	3
Endpoint ID	54242
IP	10.1.10.12
Interface	lan
Interface Role	lan
Port	33781

Like other UTM logs, newly added DNS and SSH UTM references can also be shown in the FortiAnalyzer Log View. Clicking the count next to the DNS or SSH event opens the respective UTM log.

3. Go to *FortiView > Threats > Top Threats*. All threats detected by any CSF member are shown.



4. The created UTM reference is also transparent to the FortiGate when it gets its logs from the FortiAnalyzer. On the FortiGate, the traffic log shows UTM events and referred UTM logs from other CSF members, even though the FortiGate does not generate those UTM log fields in its traffic log. In this example, the CSF child FortiGate shows the referred UTM logs from the CSF root FortiGate.

The screenshot shows the FortiGate 61E Log & Report interface. The left sidebar contains navigation options: Dashboard, Security Fabric, FortiView, Network, System, Policy & Objects, Security Profiles, VPN, User & Device, WiFi & Switch Controller, Log & Report (selected), Forward Traffic (selected), Local Traffic, Sniffer Traffic, System Events, Router Events, VPN Events, User Events, Endpoint Events, HA Events, WAN Opt. & Cache Events, and WiFi Events.

The main area displays a table of traffic logs with columns: #, Date/Time, Source, Destination, and Security Events. The table shows 22 entries, with the 4th entry highlighted in yellow. The Security Events column shows AV, WEB, and APP icons.

The right sidebar shows Log Details for the selected entry, including details for AntiVirus and Web Filter.

AntiVirus Details:

- Agent: curl/7.19.7
- FortiSandbox Checksum: 275a021bbfb
- Submitted to FortiSandbox: false
- Details: host: 172.18.32.126
- Threat Level: critical
- Device ID: FG100D3G16
- Direction: incoming
- Detection Type: Virus
- Log event original timestamp: 1547077055
- Event Type: infected
- File Name: eicar.com
- Message: File is infected
- Profile Name: default
- Quarantine Skip: File-was-not-c
- Reference: https://fortigu
- Type: utm
- URL: http://172.18.
- Virus/Botnet: EICAR_TEST_
- Virus ID: 2172

Web Filter Details:

- Category: 255
- Device ID: FG100D3G16
- Direction: outgoing
- Log event original timestamp: 1547077055
- Event Type: urlmonitor
- Hostname: 172.18.32.126

Security Fabric ADOMs

All Fortinet devices included in a Security Fabric can be placed into a Security Fabric ADOM, allowing for fast data processing and log correlation. Fabric ADOMs enable combined results to be presented in the *Device Manager*, *Log View*, *FortiView*, *Incidents & Alerts* and *Reports* panes.

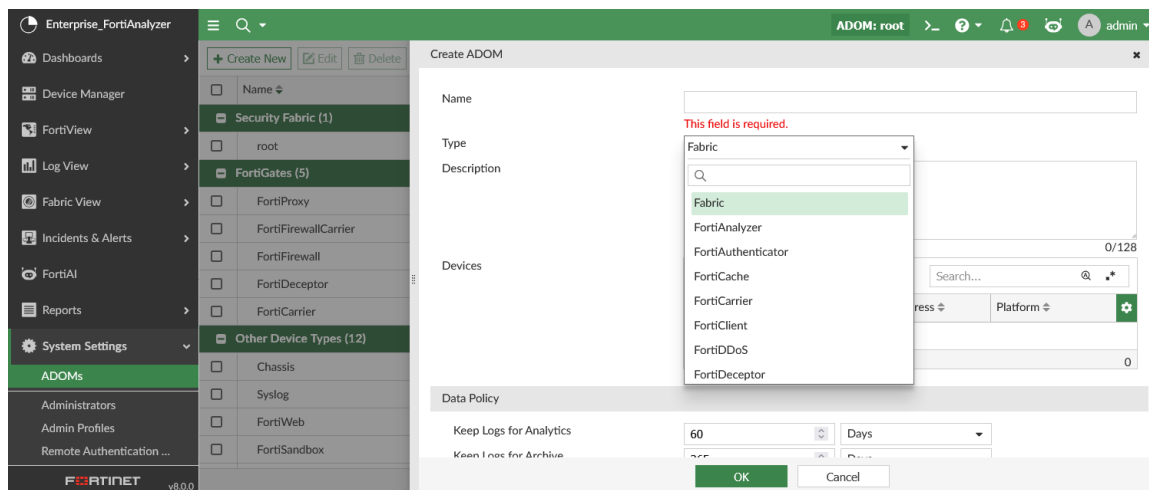
In a Fabric ADOM:

- *Device Manager*: View and add all Fortinet devices in the Security Fabric to the Fabric ADOM, including FortiGate, FortiSandbox, FortiMail, FortiDDoS, and FortiClient EMS.
- *Log View*: View logs from all Security Fabric devices.
- *FortiView*: FortiDDoS and FortiClient EMS widgets are available.
- *Incidents & Alerts*: Predefined alert handlers for FortiGate, FortiSandbox, FortiMail, and FortiWeb ADOMs are available, and triggered alerts are displayed for all device types.
- *Reports*: View predefined reports, templates, datasets, and charts for all device types. Charts from all device types can be inserted into a single report.

Creating a Security Fabric ADOM

To create a Fabric ADOM:

1. In FortiAnalyzer, go to *System Settings > ADOMs*.
2. Select *Create New*.
3. Configure the settings for the new Fabric ADOM and select *Fabric* as the type.
See [Creating ADOMs on page 539](#) for more information on the individual settings.



4. Select *OK* to create the ADOM.
The Fabric ADOM is listed under the *Security Fabric* section of *All ADOMs*.

Name	ADOM Type	Allocated Storage	Devices	Comments	Time Zone
Security Fabric (1)					
root	Fabric	32 GB	5 Devices (including 5 VDOMs)		US/Pacific (-08:00 PST)
FortiGates (5)					
FortiProxy	FortiProxy	1000 MB			US/Pacific (-08:00 PST)
FortiFirewallCarrier	FortiFirewallCarrier	1000 MB			US/Pacific (-08:00 PST)
FortiFirewall	FortiFirewall	1000 MB			US/Pacific (-08:00 PST)
FortiDeceptor	FortiFirewall	1000 MB			US/Pacific (-08:00 PST)
FortiCarrier	FortiCarrier	1000 MB			US/Pacific (-08:00 PST)
Other Device Types (12)					
Chassis	-	Unlimited			US/Pacific (-08:00 PST)
Syslog	Syslog	1000 MB			US/Pacific (-08:00 PST)
FortiWeb	FortiWeb	1000 MB			US/Pacific (-08:00 PST)
FortiSandbox	FortiSandbox	1000 MB			US/Pacific (-08:00 PST)

Migrating to a Fabric ADOM

You can change an existing non-Fabric ADOM to a Fabric ADOM using the FortiAnalyzer CLI.

1. In the FortiAnalyzer CLI, enter the following commands:

```
execute migrate fabric <fabric name>
```

A note is displayed informing you of the number of ADOMs that will be affected, and once begun, a summary is displayed and the system will reboot.

Enabling SAML authentication in a Security Fabric

When FortiGate is configured as a SAML SSO IdP in a Security Fabric, FortiAnalyzer can register itself to FortiGate as an SAML service provider, allowing for simplified configuration of SAML authentication.

When FortiAnalyzer is configured as a Fabric SP, a default SSO administrator is automatically created for each Security Fabric. When a user logs in through Fabric SSO, the Fabric IdP provides the user's profile name. If FortiAnalyzer has a profile with a matching name, the profile is assigned to the user. Otherwise, the profile of the SSO administrator is assigned to the user by default.

Before configuring FortiAnalyzer as a Fabric SP, *Security Fabric Connection* and *FortiAnalyzer Logging* must be configured on the root FortiGate.



When ADOMs are enabled, SSO users can only access the ADOM that includes the root FortiGate.

To configure FortiAnalyzer as a Fabric SP:

1. Before configuring FortiAnalyzer as a Fabric SP, configure the following on the root FortiGate:
 - a. To enable *FortiAnalyzer Logging* on the root FortiGate, see [Logging to FortiAnalyzer](#) in the FortiGate / FortiOS Administration Guide.
 - b. To configure a *Security Fabric Connection* on the root FortiGate, see [Configuring the root FortiGate and downstream FortiGates](#) in the FortiGate / FortiOS Administration Guide.

- c. To enable *SAML Single Sign-on* on the root FortiGate, see [Configuring the root FortiGate as the IdP](#) in the FortiGate / FortiOS Administration Guide.
2. In the FortiAnalyzer GUI, enable the *Fabric SP Single Sign-On Mode*.
 - a. Go to *System Settings > SAML SSO*.
 - b. Select *Fabric SP* as the *Single Sign-On Mode*.
 - c. Enter the address of the FortiAnalyzer SP.
 - d. Select a *Default Admin Profile*.

The *Auto Create Admin* setting is enabled by default for Fabric SP. SSO admins will be created automatically if they do not exist. The admins can be found in the FortiAnalyzer GUI under *System Settings > Administrators* with the name "CSF_SSO_FG<serial number>".

- e. Click *Apply*.

The FortiAnalyzer will automatically detect the IdP FortiGate and register itself as a SAML SP. This process may take up to ten minutes. Once completed, IdP information is displayed in the Fabric IdPs table on FortiAnalyzer, and SP information can be viewed in FortiOS.

FAZVM64-104

ADOM: root

admin

System Settings > SAML SSO

Single Sign-On Settings

Server Address

Allow admins to login with FortiCloud

Single Sign-On Mode: Disabled Identity Provider (IdP) Service Provider (SP) **Fabric SP**

In Fabric SP mode, an SSO administrator is created for each Security Fabric. When a user logs in via Fabric SSO, the Fabric IdP provides the user's profile name. If this system has a profile with the matching name, the profile is assigned to the user. Otherwise, the profile of the SSO administrator is assigned to the user by default.

Default Admin Profile: Standard_User

Auto Create Admin: ☒

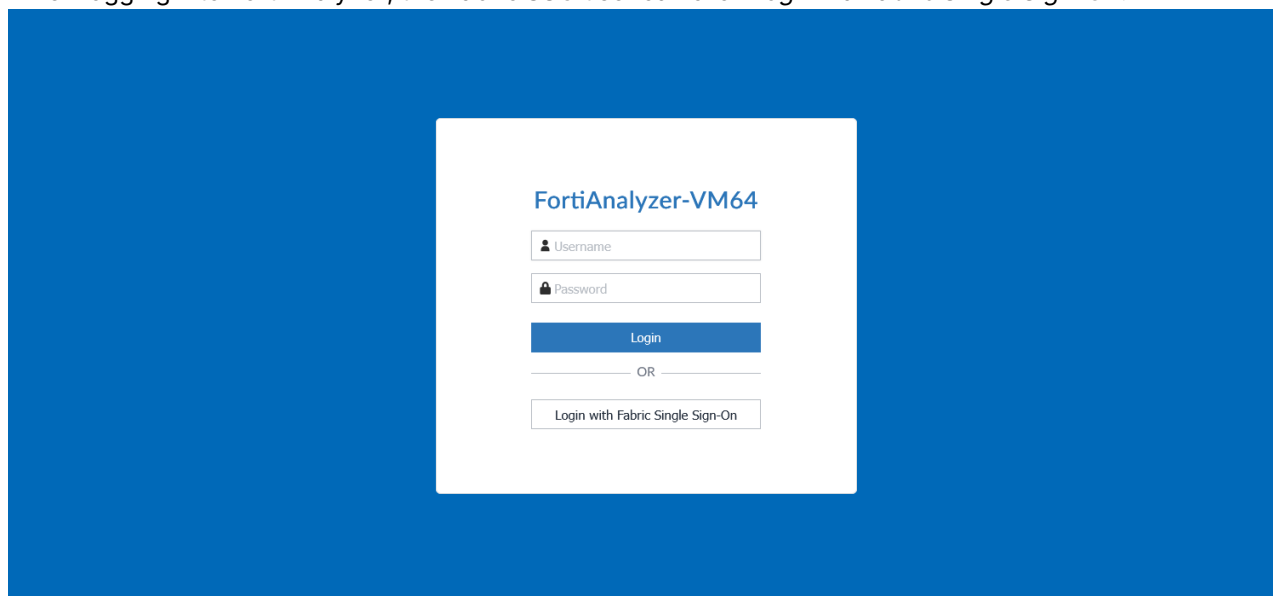
Fabric IdPs

Root Device	ADOM Name	Status	IdP Settings
☐ FGT91E4Q16	root	Enabled	Entity ID: http://.../saml-idp/cs Login URL: https://.../saml-idp/ Logout URL: https://.../saml-id
☐ FG101E4Q17	test	Enabled	Entity ID: http://.../saml-idp/ Login URL: https://.../saml-id/ Logout URL: https://.../saml-i

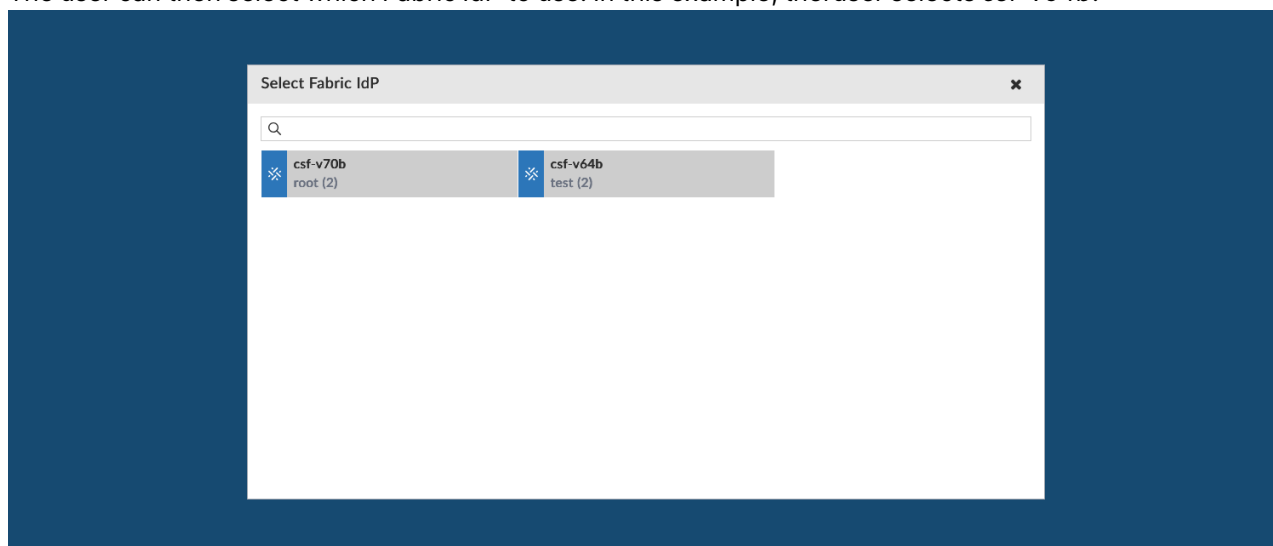
2

Apply

3. When logging into FortiAnalyzer, the Fabric SSO user can click *Login via Fabric Single Sign-On*.



The user can then select which Fabric IdP to use. In this example, the user selects *csf-v64b*.



The user is then logged into FortiAnalyzer with access according to the admin profile.

